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Code War:

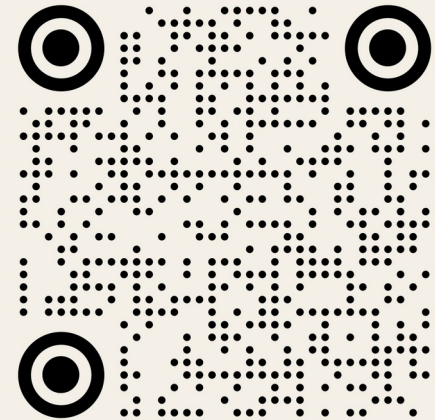
Cold War History through Game Modding

The overall classification of this brief is:
THIS NEVER HAPPENED// REL DONT TELL MOM

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jacob.j.boyce@cybertacticsforge.org

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Agenda:

WARNO:

Scope and Sequence

Rules of Engagement:

Basic Game Mechanics

Historical Perspective

Operations Overview:

Historical context

OPORD/0readme.txt

10 -20 Minute Live Example

Operation Debrief:

Infantry.yaml

Additional Units

Notes in comments

SITREP (Situation Report)

How to Implement and Next Steps



WARNING ORDER (WARNO)

Scope and Sequence

Situation

15x120min class periods

68 Students across 5 periods

Mission

Engage Historical Content
Teach Coding From Top Down

Execution

Allow for Exploration
Protect Realism

Admin/Logistics

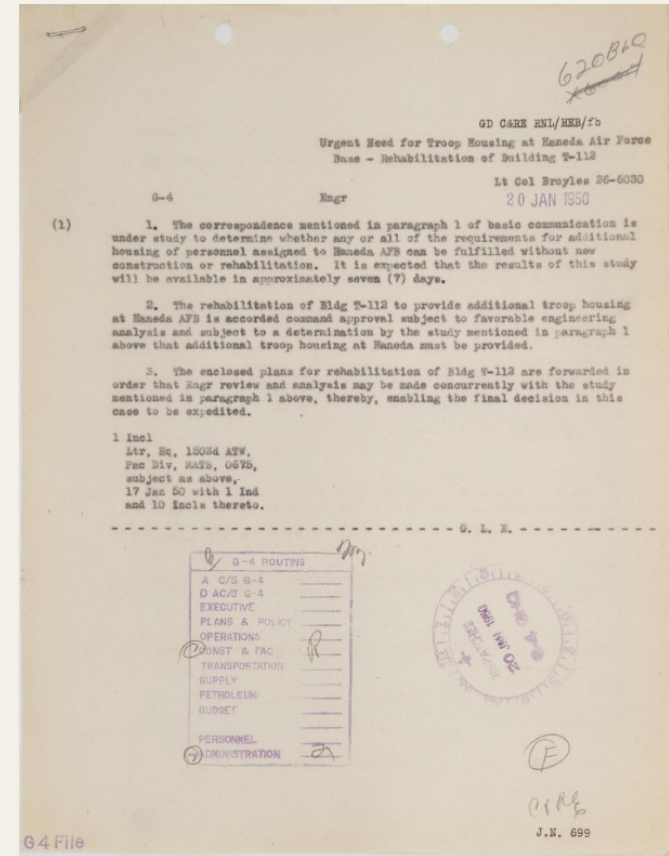
Linux Mint/Visual Studio/.net

github.com/OpenRA/OpenRAModSDK

Command/Signal

CSTA K12 Computer Science Standards

AP US History/ 20th Century History



Declassified WARNO from 1950

Rules of Engagement:

Context: Cold War "goes hot" mid-1990s

Basic Game Mechanics of Red Alert

Real time strategy

Build forces to use

Resource allocation

Balance money and power

Mining ore and building power

Balance attack, defense & capacity

Build defensive weapons

Turrets/AA-gun

Build offensive weapons

Tanks/Aircraft/Ships

Build capacity

Tech Centers/Radar

:insert video here?

Evacuation or Escalation?

General MacArthur had told General Collins that, under existing conditions, the UN Command would be unable to remain in Korea. General Collins had privately disagreed. But if General MacArthur was right, then a refusal either to send reinforcements or to alter the rules of engagement would render evacuation unavoidable. The prospect had to be squarely faced.⁶³

On 22 December the Joint Chiefs of Staff held a critically important discussion on courses of action in Korea. The occasion was General MacArthur's request, still pending, for National Guard divisions. The Joint Chiefs of Staff agreed that a decision should be made now, "on the governmental level," that US forces would be evacuated as soon as it became apparent that the Chinese Communists intended to drive the United Nations from Korea. An attack in force on UN positions, accompanied by a continuing buildup of enemy forces, would constitute sufficient evidence of such intention, and if it took place, "it must be accepted that we are being militarily forced out of Korea." In any event, the original UN directive, under which the United States had gone into Korea, was now obsolete and should be revised. They directed the Joint Strategic Survey Committee to prepare a memorandum for the Secretary of Defense that would serve to bring these issues to the attention of the National Security Council. Also, General MacArthur was informed that, pending decisions on the above matters, no additional divisions would be deployed to the Far East.⁶⁴

Before the Committee had replied to this directive, the President summoned Secretaries Acheson and Marshall and General Bradley to Blair House on 26 December, to discuss the Korean situation and the possibility of withdrawal. Secretary Acheson stressed that the US objective, resisting aggression "on a collective security basis" for as long as possible, was very clear and was accepted by all present. Reverses, he continued, did not warrant withdrawal; the UN Command was not "hopelessly outnumbered" and it enjoyed the inherent advantage

GEN MacArthur addresses strategic capability under current RoE in the face of Chinese Communist forces in Korean Peninsula

Rules of Engagement:

Historical Perspective

Cold War History

Students are put in position of LTC
(05) strategic operations officers

War-simulation technology:

Air Defense Simulation (1948)

Carmonette Series (1953)

Janus (Late 1970s)

RAND Strategy Assessment System (1980s)

Focus on research-based accuracy

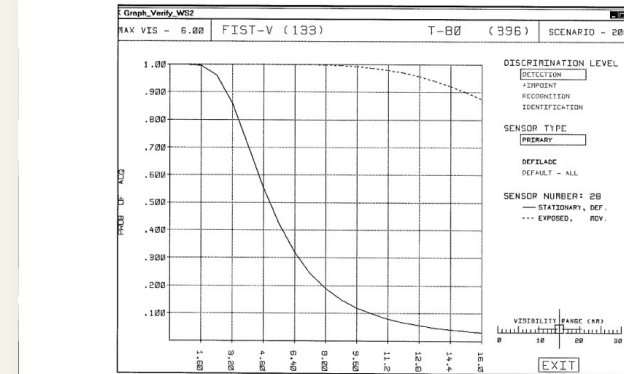
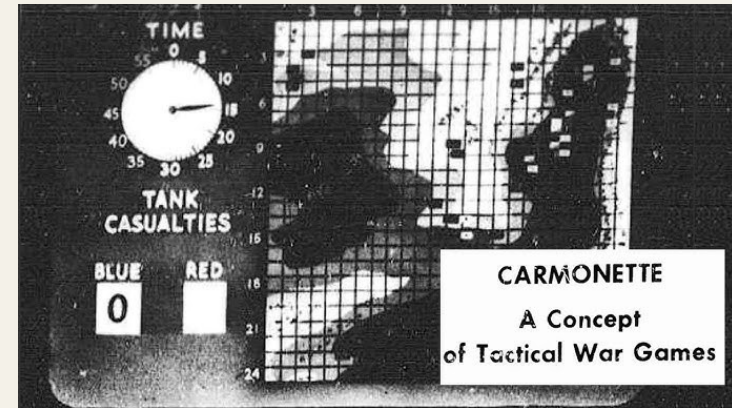


Figure 1. Probability of detection curve versus range (km) for a FistV w/ thermal sights vs. a stationary (solid line) T80.

Early Cold War examples of how algorithms informed decision-makers, a concept that continues to this day

Operations Overview:

Historical context for each milestone:

Lecture/Content Delivery

Contextual overview "Behind the Times"

Declassified Primary Sources

Video/Multimedia

Situation Report (SITREP) to GEN Bloom
-Check on learning

Operation (coding modding) task

Code War Site

Code War: Cold War History through Game Modding

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Code War Classification Levels

Course Modules:

1948- Preparing for Winter

- **Focus:** The Yalta and Potsdam Conferences and their role in shaping the post-WWII world order
- **Skill:** Red Alert basic gameplay, strategy, and historical modding
- **Activity:** OPERATION US VS THEM

1949- Cold Shoulder

- **Focus:** The Berlin Blockade and Berlin Airlift
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- **Focus:** Espionage and sedition: the origins and rise of cold-war era paranoia
- **Skill:** Trait manipulation in C# and YAML
- **Activity:** OPERATION SHADOW OF THE CURTAIN

jboyce1.github.io/CodeWar

Operations Overview:

Operation Order (OPORD)

Defines Objectives for students

Gives strategic reasoning

Describes scope of work expected

Defines left/right limits of simulation

Describes "deliverables" for students

Describes evaluation criteria

Additional security classification

This is an assignment without breaking character

HEADQUARTERS

STRATEGIC EXERCISE DIRECTIVE

TITLE: Comparative Analysis and Operational Effectiveness Assessment of U.S. Rifle Companies vs Soviet Rifle Companies - 1948

DATE: JANUARY 8, 1948

TO: ALL ASSIGNED PERSONNEL

FROM: LT. COL. J. J. BOYCE, STRATEGIC PLANNING DIVISION

I. OBJECTIVE:

To evaluate and compare the operational capabilities of U.S. and Soviet Rifle Companies within the context of 1948 through simulated wargaming scenarios in the Command and Conquer: Red Alert 1948 mod. The exercise aims to identify relative strengths, weaknesses, and recommended force adjustments to ensure combat superiority.

II. BACKGROUND:

Tensions along the Iron Curtain in 1948 necessitate a comprehensive analysis of NATO and Warsaw Pact forces. Both sides rely heavily on light infantry, light tank support, light artillery, and light close air support. This exercise will simulate engagements using OpenRA's 1948 mod to determine likely outcomes and highlight areas for strategic improvements.

III. SCOPE:

Participants will conduct a tactical assessment of U.S. Rifle Companies and Soviet Rifle Companies. Analysis will include:

- Unit Composition and Equipment
- Firepower and Mobility
- Defensive and Offensive Capabilities
- Support Assets Integration
- Engagement Outcomes and Tactical Adjustments

IV. SIMULATION PARAMETERS:

1. Forces and Unit Composition
 - a. U.S. Rifle Company
 - i. Platoons equipped with M1 Garand rifles, M35 trucks, and light mortars.



Operations Overview:

Oreadme.txt files:

Placed in relevant directories

Gives directions (Top)
-Make and launch mod

Gives explanations (Bottom)
-Overview
-Purpose
-Examples

Oreadme.txt files

```
CodeWar1948-usvthem-readme.txt x
Put this CodeWar1948-usvthem in the CodeWar1948 > mods > CodeWar1948 folder

Delete the old CodeWar1948 folder

Rename this CodeWar1948-usvthem to just CodeWar1948

from the root directory (CodeWar1948) run: make
from the root directory (CodeWar1948) run: ./launch-game.sh

Launch a server from the multiplayer menu and wait for others to join.
In Options: Unclick "Short Game" and from the dropdown for Starting Units, select "OP VS THEM"

Develop Teams:
Select ONLY USA and Russia factions
USA select team 1
Russia select team 2

Each USA spawns 1 Rifle Company
Each Russia spawns 2 Rifle Companies
```

```
CodeWar1948-usvthem-readme.txt x  Oreadme-rules.txt x

# RULES DIRECTORY
This directory contains configuration files that define how sprites behave and interact
with the game environment. These files combine visual elements with gameplay logic.

## Purpose of Rules Files
- Assign traits to sprites (e.g., health, speed, attack power).
- Define parameters for those traits (e.g., range, cooldown, durability).
- Specify interactions between sprites (e.g., damage types, movement restrictions).

## Examples of Rules Files
1. 'tank_rules.yaml'
- Assigns traits to the tank sprite, such as:
  - Health: 100
  - Speed: 3 (units per second)
  - Attack Power: 20
- Specifies parameters for firing:
  - Range: 10 units
  - Cooldown: 2 seconds.
2. 'helicopter_rules.yaml'
- Assigns traits to the helicopter sprite:
  - Speed: 5 (units per second)
  - Movement Type: Air
  - Armor: Light
- Defines unique behaviors, like hovering.
3. 'explosion_rules.yaml'
- Specifies the effects of the explosion animation:
  - Damage Radius: 5 units
  - Damage: 50
  - Duration: 1.5 seconds.

## Key Sections in a Rules File
- **Traits**: Define the fundamental attributes of a sprite.
- **Behavior**: Specify how the sprite interacts with other entities and the environment.
- **Parameters**: Provide detailed configurations for traits and abilities.

## Notes
- Rules files are critical for balancing gameplay. Test changes thoroughly to ensure
  fairness and functionality.
- Consistent naming conventions help organize and maintain files.
```


10 -20 Minute Live Example:

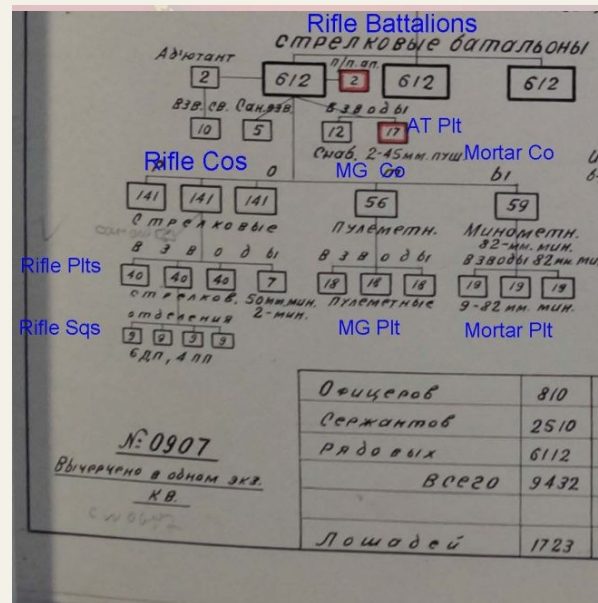
adjust infantry speed
make
launch

Operation Debrief:

Infantry.yaml:

17 unique soldier types based primarily on

TO&E 7-17 1945 & WTaT 05/40 DEC1944-49



1948 Operation Us vs Them
US vs Soviet Infantry

#US Infantry 1948	#Soviet Infantry 1948
E1_M1903A4_SNIPER: #USA BASED SNI	E1_MOSIN-NAGENT: #This is the sta battleorder.org/ussr-rifle-co-194
e1_garand_gunner: #Genevieve - Tu https://en.wikipedia.org/wiki/M1 featured/rough-guide-costs-of-wwi the primary infantry of the Unite	E1_MOSIN-NAGENT-SNIPER: #This is battleorder.org/ussr-rifle-co-194
e1_m1918a2bar_gunner: #M1918 Brow	E1_PPS: #This is the standard sub battleorder.org/ussr-rifle-co-194
Colt_M1911_Infantry: # Berg, mond	E1_DP-27: #This is the standard T www.battleorder.org/ussr-rifle-co
M2_Flamethrower: #Ezra Adelman, M marine-corps-flamethrower-tank-wc	E1_DP-27: #This is the standard T www.battleorder.org/ussr-rifle-co
E1_M1919A4_machinegunner: #Adds M	E1_SG-43: #This is the standard M www.battleorder.org/ussr-rifle-co
E1_60mm_mortar: #IOW TO&E 7-17 19	E1_50mm_mortar: #This is the sta battleorder.org/ussr-rifle-co-194
E1_M18_57mm: #IOW TO&E 7-17 1945	
SPY_USA_OSS: #Operates same as sp	
e1_m7a1_grenadier: #Adjusts infan us-army-rifle-co-1945) ...	

Operation Debrief:

14 Small Caliber

```

1  #US Small Caliber Weapons
2  > M1919A4: #usa weapon 1948 IOW
27
28 > M2_Air_To_Air: # ahn man monda
48
49 > M1911: #berg norwood monday 20
62
63 > M1918A2_BAR: #usa Browning Aut
81
82 > M2_Ariel: #.50cal Browning M2
97
98 > M1Garand: #Genevieve Sweterlit
132
133 > M1903A4_SNIPER-RIFLE: #cassius
142
143 #Soviet Small Caliber Weapons
144
145 > MOSIN-NAGENT: #Soviet Rifle Ir
164
165 > MOSIN-NAGENT_SNIPER-RIFLE: #Sa
174
175 > PPS: #Soviet submachine-gun (7
194
195 > DP-27: #Soviet light machine-g
214
215 > SG-43: #Soviet light machine-g
234
235 > VYa-23: #Soviet Cannons mounte

```

10 Artillery

```

1  #NATO
2  #USA Ballistics 1948
3  > 60mm: #usa mortar weapon in 1948
25
26 > M18_57mm: #usa mortar weapon in
48
49 > 105mm: #USA artillery weapon in
71
72 > 155mm_M1: #USA artillery weapon
95
96 > 75-mm_field_gun_M1897: #Very lar
115
116 > 76mm: ...
121
122 #Soviet
123 #Soviet Ballistics 1948
124 > 50mm: #soviet mortar weapon in 1
146
147 > 122mm: #soviet artillery weapon
169
170 > 155.4mm_ML205: #Soviet artillery
192
193 > 85mm: #Oliver Jacobs Tuesday 202

```

8 Aircraft / 3 Bombs

```

2  #USA 1948
3
4  > A-26B: # Rory Magaro Friday 2025
88
89 > B-24: #USA Liberator Medium Rang
173
174 > F-86A_Sabre: #ahn man monday 202
253
254 > P-47: #USA P-47 Thunderbolt ligh
338
339 > C-47: #USA Cargo 3 ton capacity,
399
400
401 #Russia 1948
402 > PE-2: #Soviet Petlyakov Medium R
486
487 > IL-2: #Ilyushin Il-2 Soviet ligh
571
572 > SU-27: #SU-27 developed 1983 for

```

```

1  #NATO
2  #US bombs 1948
3  > M65: #Standard 100-3
35
36 > Atomic: ...
142
143 #Soviet
144 #Soviet bombs 1948
145 > FAB-500: #1100lb hea
177

```

20 VICs

```

2  #US 1948
3
4  > USA_RIFLE_SQUAD_TRUCK: #Table of Organizati
44
45 > USA_LIGHT_MACHINE_GUN_SECTION_TRUCK: #Table
85
86 > USA_MORTAR_SECTION_TRUCK: #Table of Organiz
126
127 > USA_RECOILLESS_RIFLE_SECTION_TRUCK: #Table
167
168 > M35: #Table of Organization No. 7-17N "Inf
196
197 > G-503_WILLYS_JEEP: #Table of Organization M
248
249 > M7_PRIEST: #USA Self Propelled Arillery 105
298
299 > M40_GMC: #USA Self Propelled Arillery 155mm
348
349 > 75mm_field-gun_M1897_on_M2_carriage: #Made
395
396 > M4A376W_HVSS_Sherman: #Gabriel Stagno-thu
446
447 > M26_PERISHING: ...
497
498 #Soviet vehicles
499 #Soviet 1948
500
501 > T54: #muhammad friday 2025...
551
552 > SOVIET_US6_RIFLE_PLATOON: #https://www.batt
583
584 > SOVIET_US6_SUBMACHINEGUN_PLATOON: #https://
613
614 > SOVIET_GAZ67_MACHINEGUN_SECTION: #https://v
666
667 > SOVIET_GAZ67_50MM_MORTAR_SECTION: #https://
719
720 > GAZ67: #Soviet Jeep https://www.battleorder
771
772 > SU-122_HOWITZER: #Soviet Self Propelled Ari
821
822 > ISU-152: #Soviet Self Propelled Arillery 12
871
872 > M1939: #Oliver Jacobs, Tuesday 2025 Soviet

```

Operation Debrief:

Research notes in comments:

BuildPaletteOrder: 10 #I made this especially small because the Garand Gunner was a common type of soldier. Therefore, it should be an easily available soldier, and should be seen at the beginning so it can act as a fighter before others can be produced.

Mobile:

Speed: 34 #Because the garand guns were heavy and would have hindered movement, I decided to give these infantry a lower speed than some other infantry.

Valued:

Cost: 40 #Cost reduced to negligible based on conscription in Soviet Union

AmmoPool@MOSIN-NAGENT:

Name: MOSIN-NAGENT_ammopool

Ammo: 70 #40 rounds (ready)8x Clips (5 rd) held in 4 leather pouches mounted on the front of belt, 30 rounds (reserve) 6x Clips (5 rd) held in 1 canvas reserve pouch on center back of belt

M1919A4: #usa weapon 1948 IOW Table of Organization No. 7-17N "Infantry Rifle Company" published 16 April 1948

ReloadDelay: 5

Range: 9c0 #adjusted to give additional effective range over M1Carbine

Inaccuracy: 191 #increased inaccuracy based on effective range and the PPS being made as a low-cost personal defense weapon

Situation Report (SITREP) :

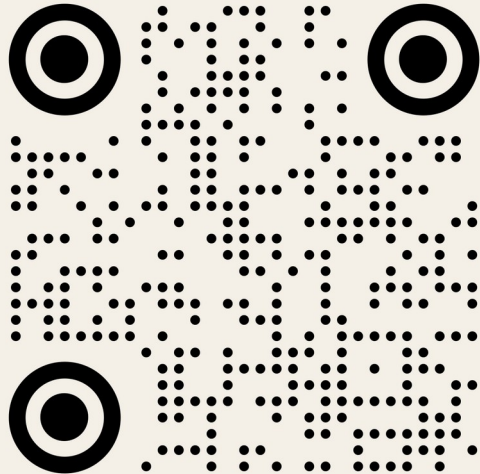
Year 1: Class progress through 1949

Year 2: Pickup where year 1 left off

Year 3 +: Continue Process

Discord server for those committed:

<https://discord.gg/uxtHTRdw>



J. Boyce user: Werjun

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jboyce1.github.io/CodeWar

Congratulations, you volunteered.

Grab your kit.

Grab a battle buddy.

Report to the woodline.

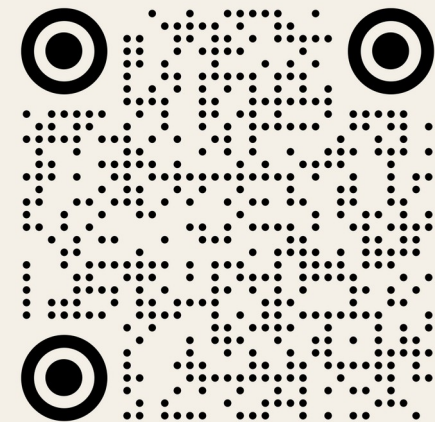
Lock and load.

Welcome to the suck.

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Let us know how we did!

After the conference, check your e-mail for a link from Survey Monkey to share your feedback!

Conference surveys must be completed in order to fulfill the requirements of Travel Reimbursement Grants.

Contact info@cyber.org if you don't receive the link within 24hrs.



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